

Software Requirements Specification (SRS)

Project: Digital Canteen Management System (DCMS)

1. Introduction

The Digital Canteen Management System (DCMS) is a web-based application that digitizes the entire canteen ordering process for students, faculty, canteen staff, and administrators. It allows users to browse the menu, place and pay for orders online, track order status and queue position in real time, and collect their food using a QR-code-based pickup system. The system replaces manual, queue-based ordering with a transparent, role-based digital workflow.

2. Objectives

- Seamless online menu browsing and ordering
- Multiple payment options (online and cash)
- Real-time order tracking and queue management
- QR-code based order verification and pickup
- Role-based access control for Students, Faculty, Canteen Staff, and Admin
- Inventory and stock management for canteen staff
- Transparent feedback, ratings, and complaint resolution
- Loyalty points and coupon-based engagement

3. Users

Student, Faculty, Canteen Staff, Administrator

4. Functional Requirements

- User registration/login with JWT authentication and role-based access
- Browse, search, and filter menu items
- Cart management and checkout
- Online (Razorpay) and cash payments
- Queue/token generation and order placement
- Real-time order tracking and QR-based pickup
- Multi-counter order processing
- Order history and re-order
- Faculty priority, scheduled, and bulk orders
- Inventory, menu, category, and pricing management
- Coupons, loyalty points, complaints, and feedback

- Notifications (Email/SMS/Push)
- Admin dashboard with reports, analytics, and announcements

5. Non-Functional Requirements

- Secure authentication using Spring Security, JWT, and BCrypt
- Fast REST API performance
- Responsive React.js interface
- Reliable order and transaction auditing
- Modular and maintainable Spring Boot architecture
- Scalable to support concurrent multi-counter operations during peak hours.

6. Technology Stack

Frontend: React.js

Backend: Spring Boot (REST APIs)

Database: PostgreSQL

ORM: Spring Data JPA / Hibernate

Security: Spring Security + JWT

External Services: Razorpay (Payment Gateway), JavaMail/SendGrid (Email), Twilio/MSG91 (SMS), Cloudinary (Image Upload), ZXing (QR Code Generator)

IDE: IntelliJ IDEA / VS Code

7. Database Tables

users, roles, students, faculty, canteen_staff, admin, menu_items, categories, cart, cart_items, orders, order_items, queue, payments, qr_codes, counters, inventory, coupons, reviews, complaints, notifications, announcements

8. Future Enhancements

- AI-based food recommendation engine based on order history
- Peak hour prediction and smart queue allocation
- Popular combination and smart search suggestions
- Sentiment analysis on reviews and feedback
- Cashless/wallet-based payment integration
- Mobile application for Android and iOS